

Executive Summary

Customer Churn Prediction & Risk Segmentation

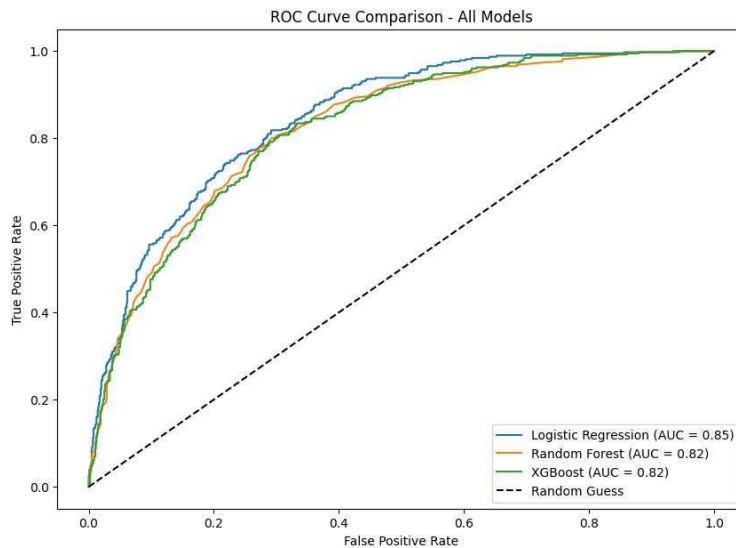
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1. Project Overview

The goal of this analysis was to build an end-to-end machine learning pipeline to predict customer churn for a telecom service provider. By identifying at-risk customers before they leave, the business can implement proactive retention strategies to protect revenue.

2. Model Performance & Selection

After training and comparing three classification models—Logistic Regression, Random Forest, and XG Boost—Logistic Regression was selected as the best performing model.



- **Best Metric:** Achieved an ROC-AUC of 0.85, indicating strong ability to distinguish between churners and loyal customers.
- **Evaluation:** The model provides a balanced approach to precision and recall, ensuring that we capture most churners without excessive "false alarms."

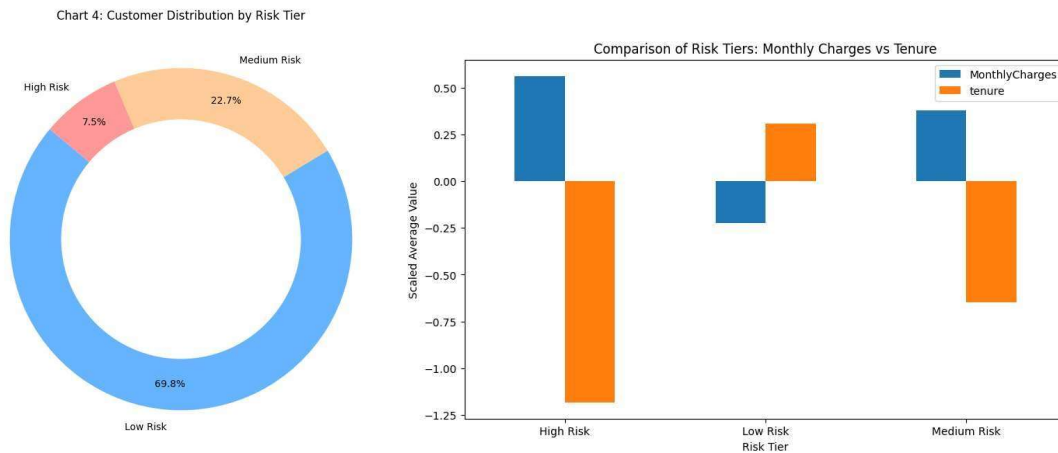
3. Key Drivers of Churn

Based on the feature importance analysis, the top three factors driving customer churn are:

1. Contract Type: Customers on "Month-to-Month" contracts have a significantly higher churn rate compared to those on long-term contracts.
2. Internet Service (Fiber Optic): There is a strong correlation between churn and Fiber Optic service, suggesting potential issues with pricing or service satisfaction in this segment.
3. Tenure: Customers in their first 6–12 months of service are at the highest risk of leaving.

4. Customer Risk Segmentation

Using the model's predicted probabilities, customers were segmented into three tiers:



- 🔴 High Risk (Prob ≥ 0.70): Customers likely to leave immediately; typically characterized by low tenure and high monthly charges.
- 🟠 Medium Risk (Prob 0.40–0.69): Customers showing early warning signs of dissatisfaction.
- 🟢 Low Risk (Prob < 0.40): The loyal customer base with high tenure and long-term contracts.

5. Actionable Business Recommendations

To reduce churn, the company should implement the following strategies:

- Targeted Contract Incentives: Offer high-risk "Month-to-Month" customers a specialized discount or value-add (like a free speed upgrade) if they switch to a 1-year or 2-year contract.
- Proactive Onboarding Support: Launch a "New Customer Success" program for Fiber Optic users in their first 6 months to address technical issues or billing concerns proactively before they lead to cancellation.

6. Limitations & Future Improvements

The current model is limited by its reliance on historical data and does not include real-time sentiment from customer support logs. Future improvements could involve integrating real-time usage patterns and competitor pricing data to increase predictive accuracy.